

rxfgnboe5

February 5, 2026

```
[2]: # pip install seaborn
```

```
[3]: import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

```
[4]: df= pd.read_csv("data.csv")
```

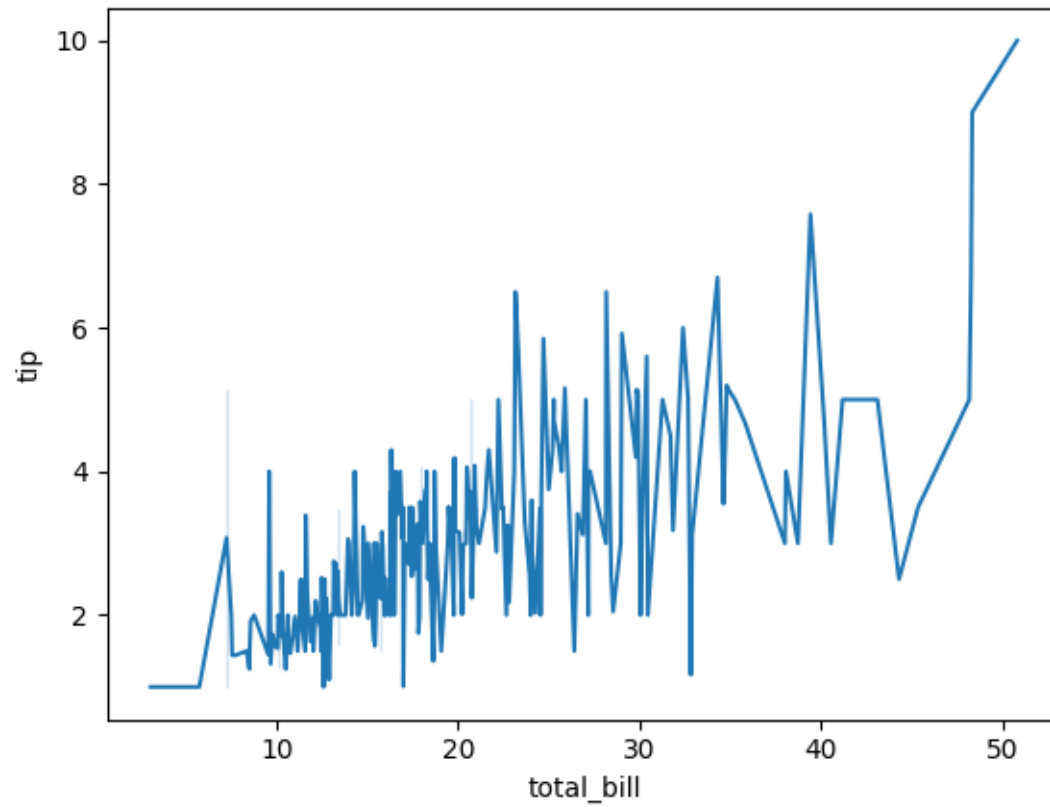
```
[5]: df
```

```
[5]:
```

	total_bill	tip	gender	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
..	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

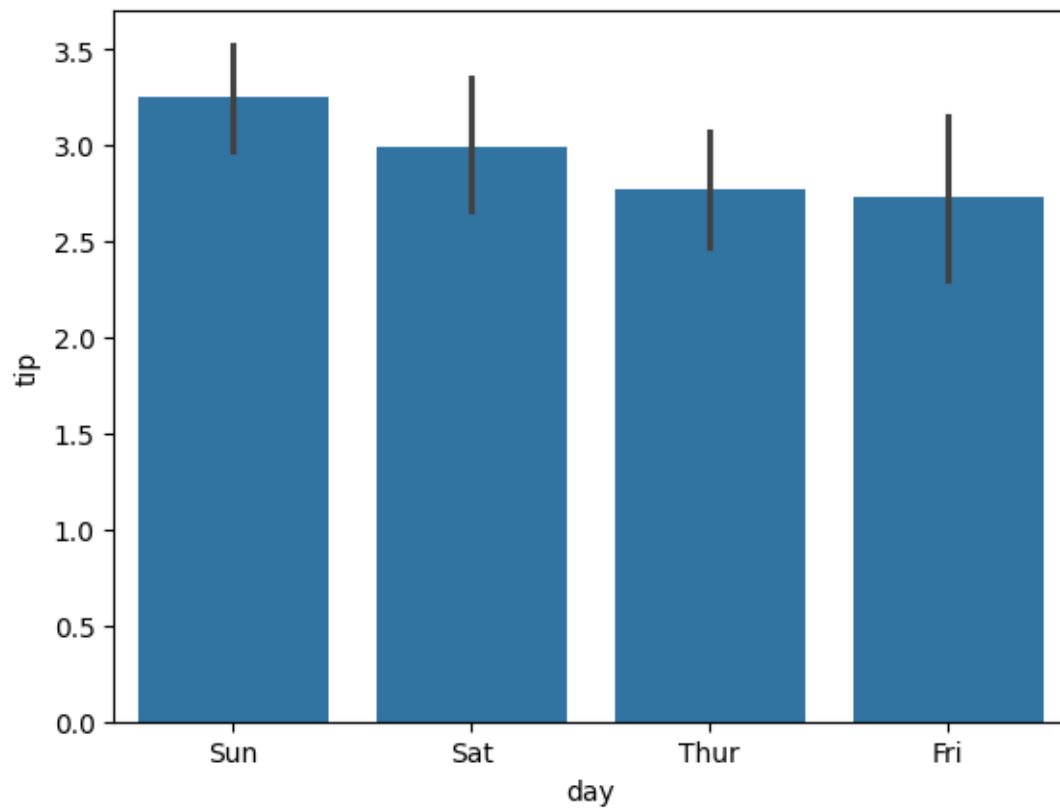
[244 rows x 7 columns]

```
[7]: sns.lineplot(x="total_bill",y="tip", data=df)
plt.show()
```



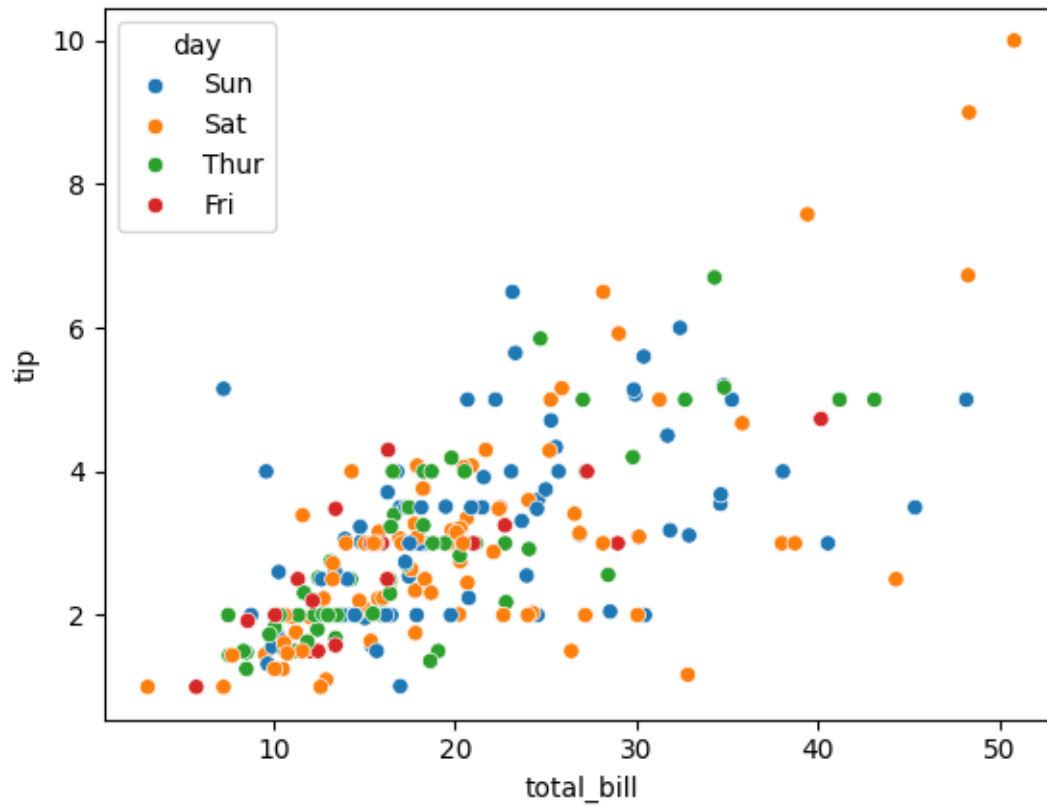
```
[8]: sns.barplot(x="day", y="tip", data=df)
```

```
[8]: <Axes: xlabel='day', ylabel='tip'>
```



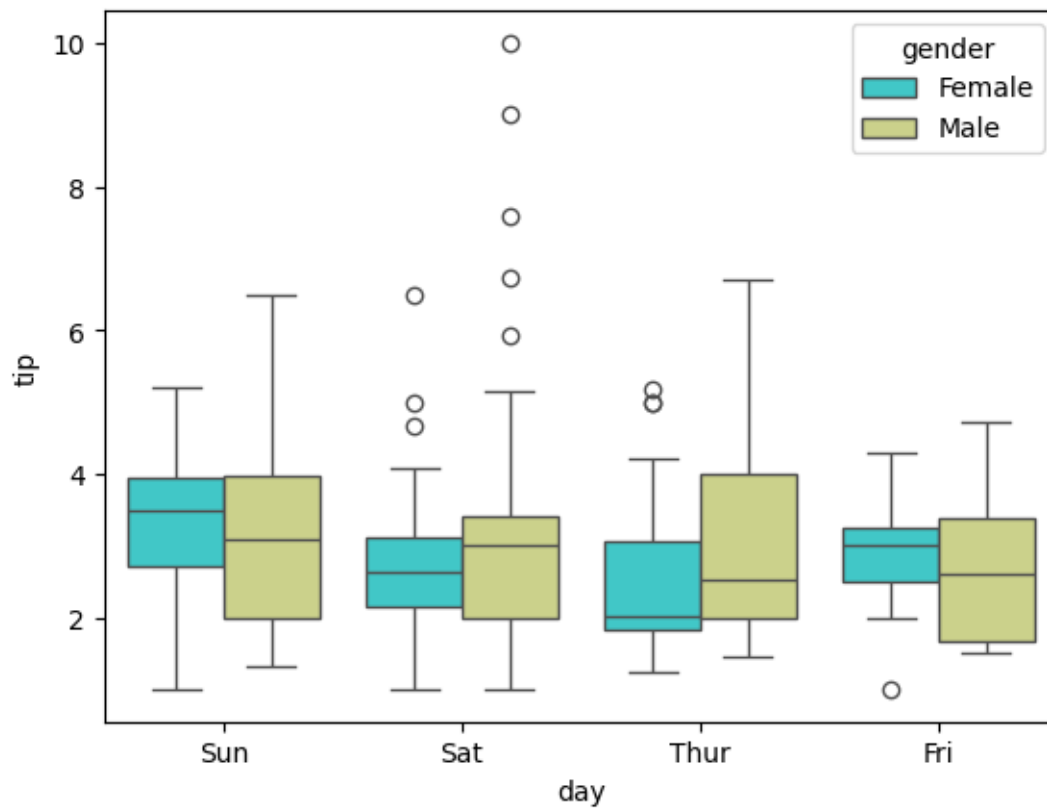
```
[11]: sns.scatterplot(x="total_bill", y="tip", hue="day", data=df)
```

```
[11]: <Axes: xlabel='total_bill', ylabel='tip'>
```



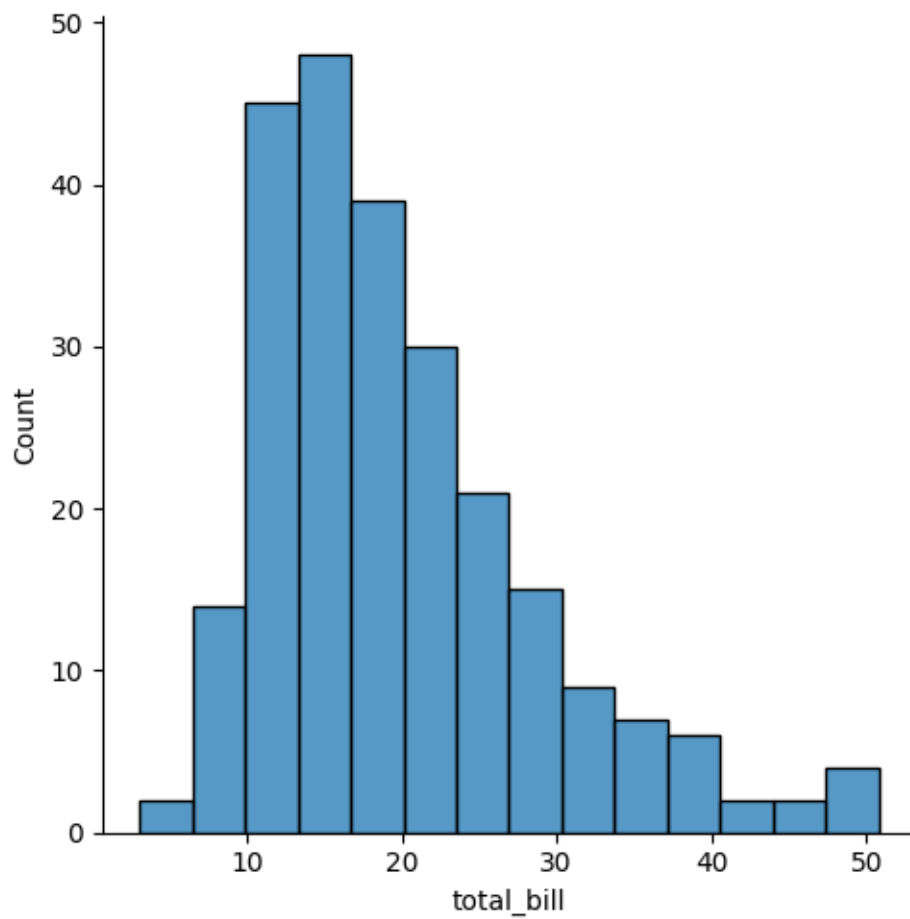
```
[16]: sns.boxplot(x="day", y="tip", data=df, palette="rainbow", hue="gender")
```

```
[16]: <Axes: xlabel='day', ylabel='tip'>
```



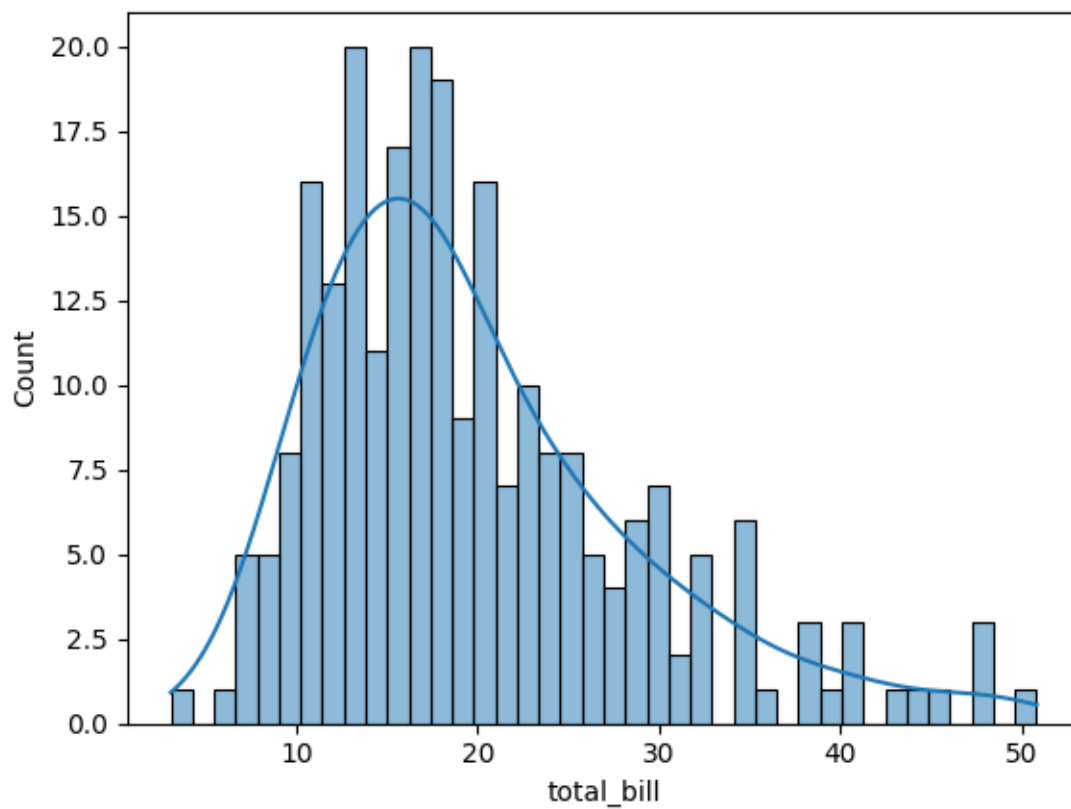
```
[17]: sns.displot(x="total_bill", data= df)
```

```
[17]: <seaborn.axisgrid.FacetGrid at 0x1ed7f94bd10>
```



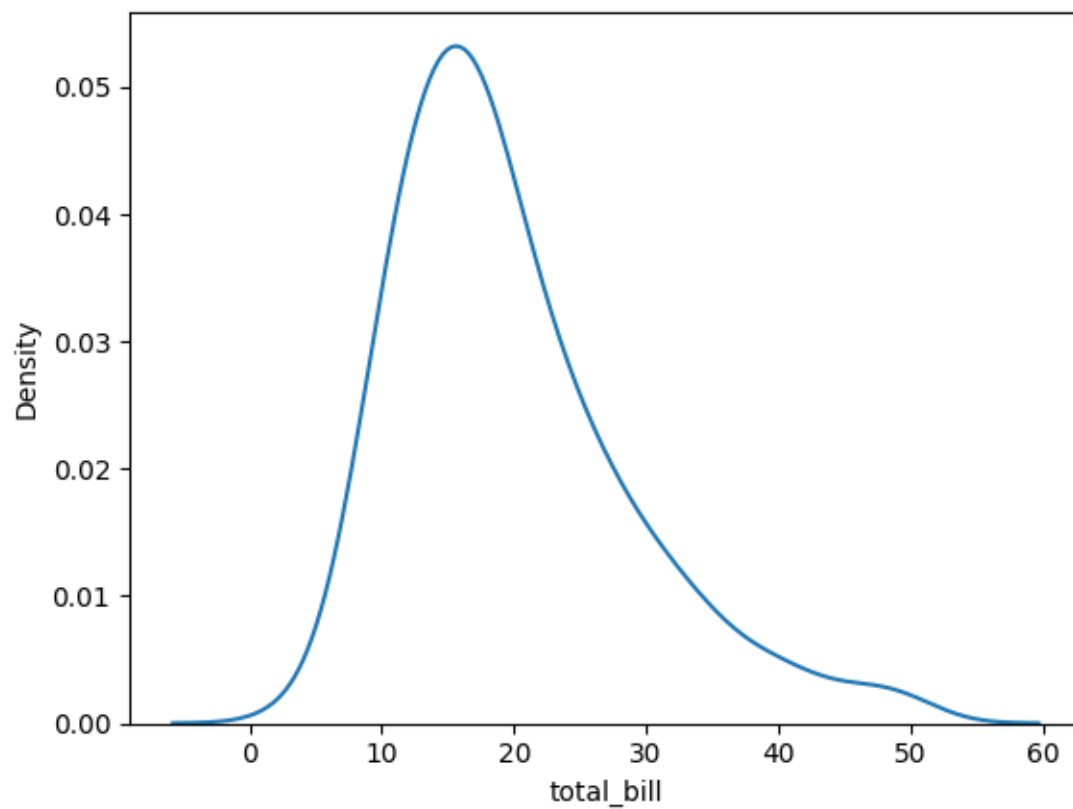
```
[21]: sns.histplot(df.total_bill, kde=True, bins=40)
```

```
[21]: <Axes: xlabel='total_bill', ylabel='Count'>
```



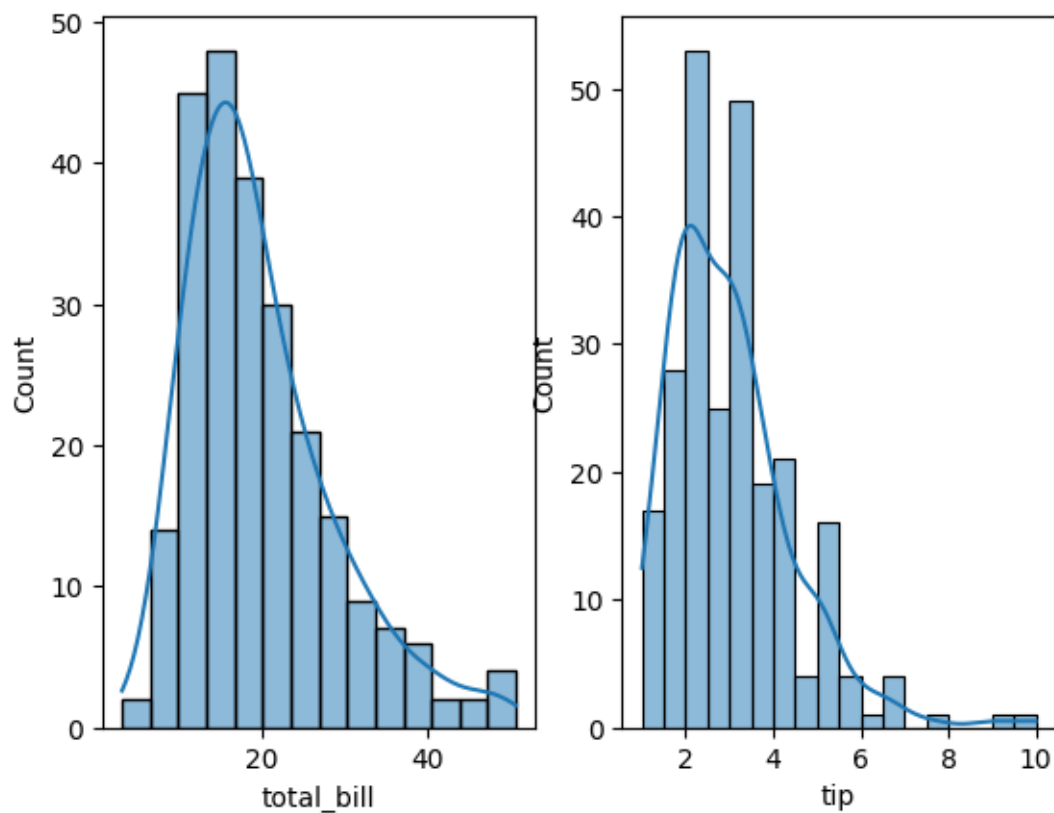
```
[22]: sns.kdeplot(df.total_bill)
```

```
[22]: <Axes: xlabel='total_bill', ylabel='Density'>
```



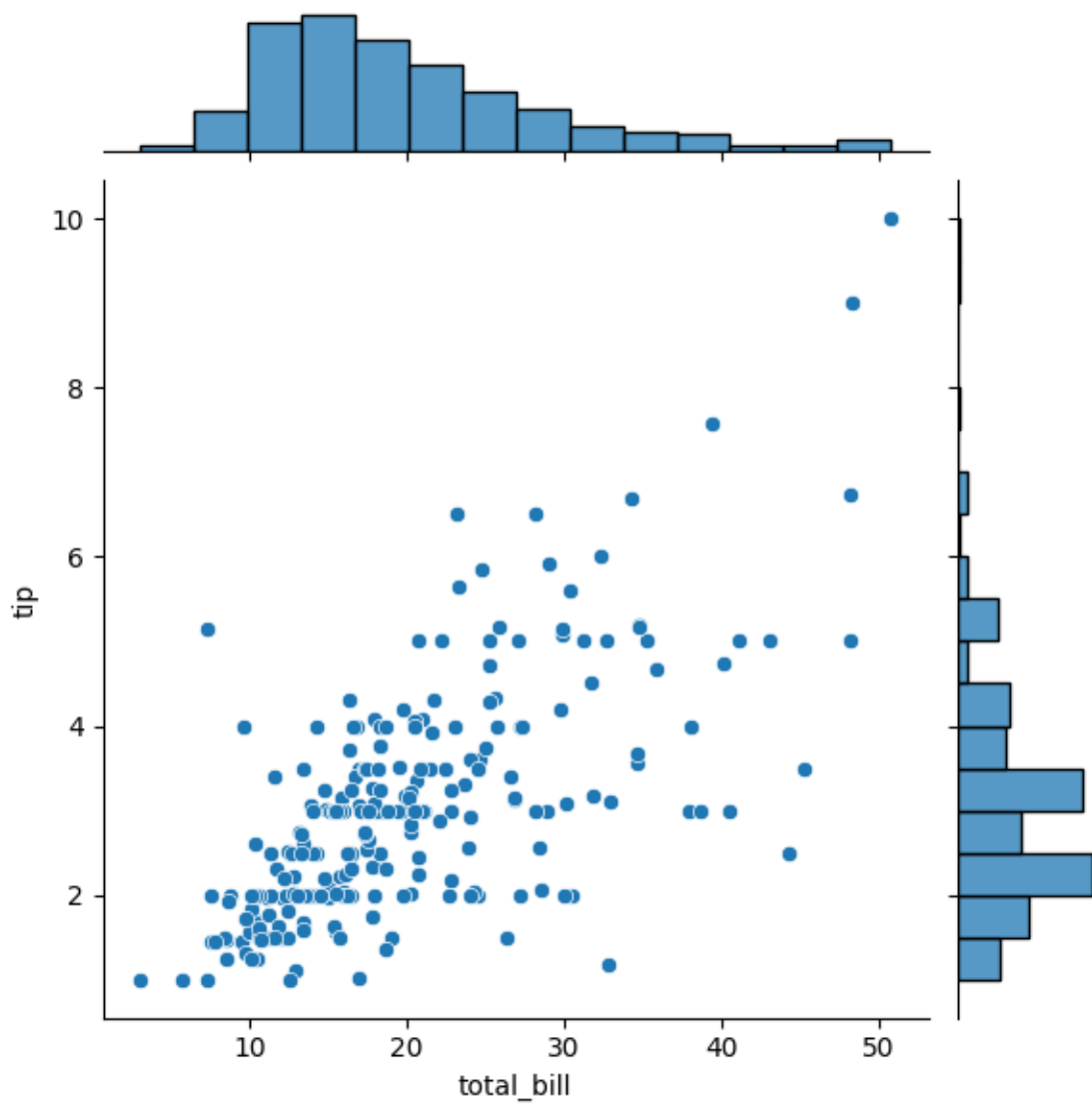
```
[25]: plt.subplot(1,2,1)
      sns.histplot(df.total_bill, kde= True)
      plt.subplot(1,2,2)
      sns.histplot(df.tip, kde= True)
```

```
[25]: <Axes: xlabel='tip', ylabel='Count'>
```

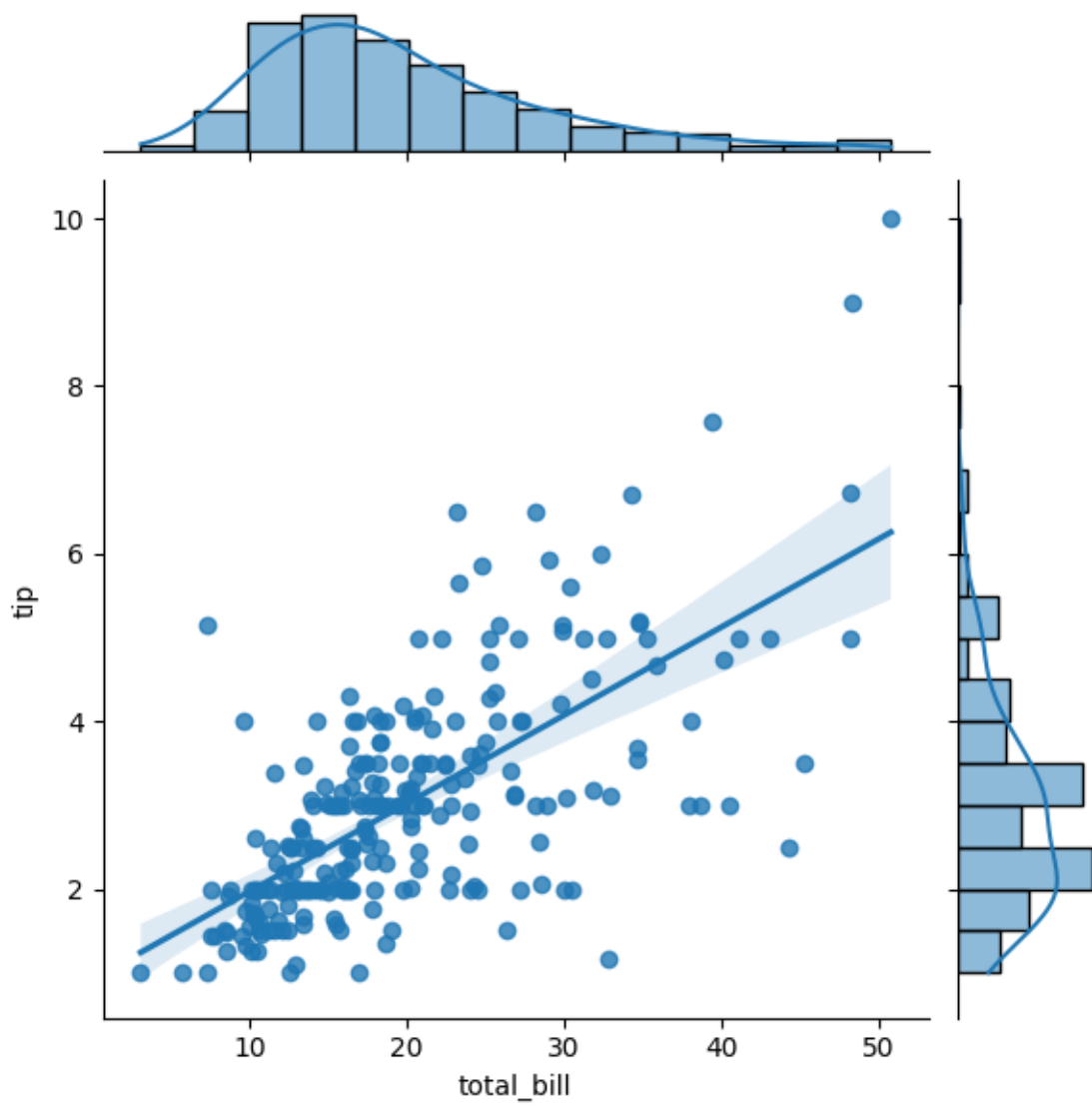
```
[27]: sns.jointplot(x="total_bill", y="tip", data=df, kind="scatter")
```

```
[27]: <seaborn.axisgrid.JointGrid at 0x1ed066bf610>
```



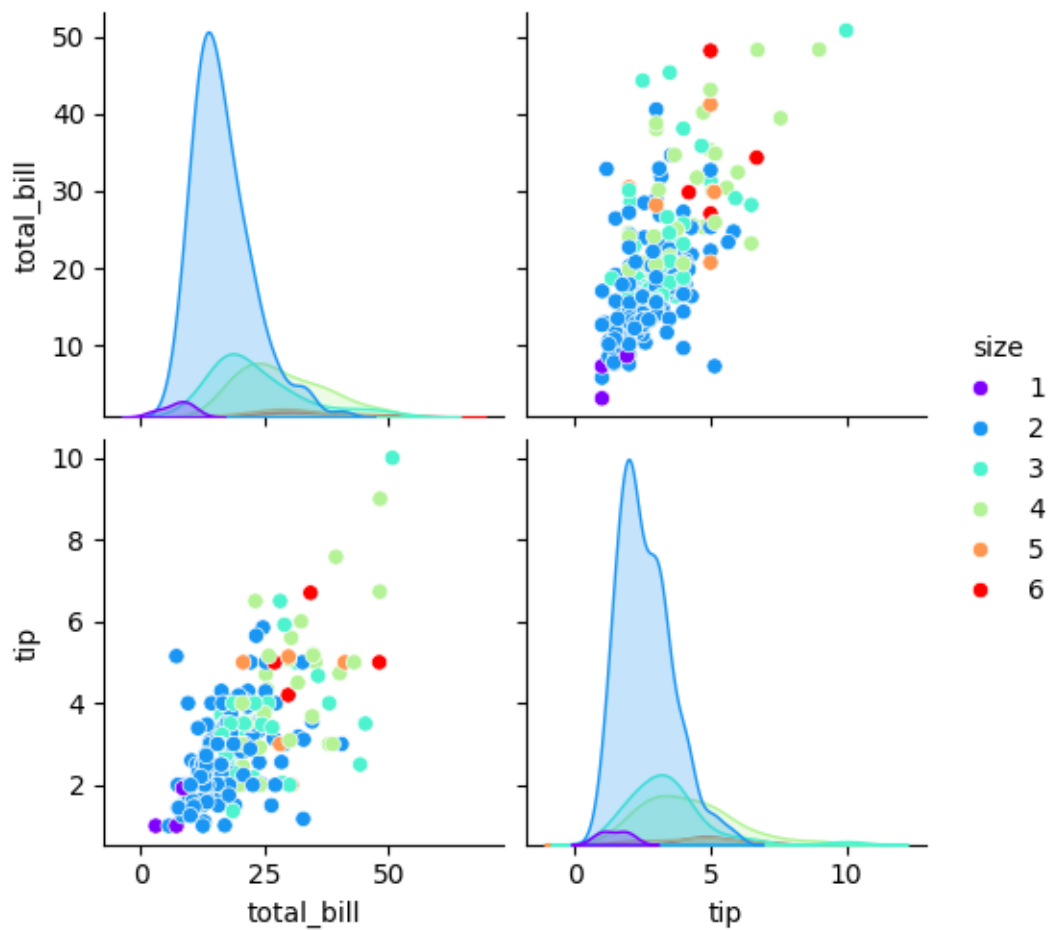
```
[28]: sns.jointplot(x="total_bill", y="tip", data=df, kind="reg")
```

```
[28]: <seaborn.axisgrid.JointGrid at 0x1ed07468810>
```



```
[34]: sns.pairplot(df, hue="size", palette="rainbow")
```

```
[34]: <seaborn.axisgrid.PairGrid at 0x1ed0ac652d0>
```



```
[37]: c= df.corr(numeric_only=True)
```

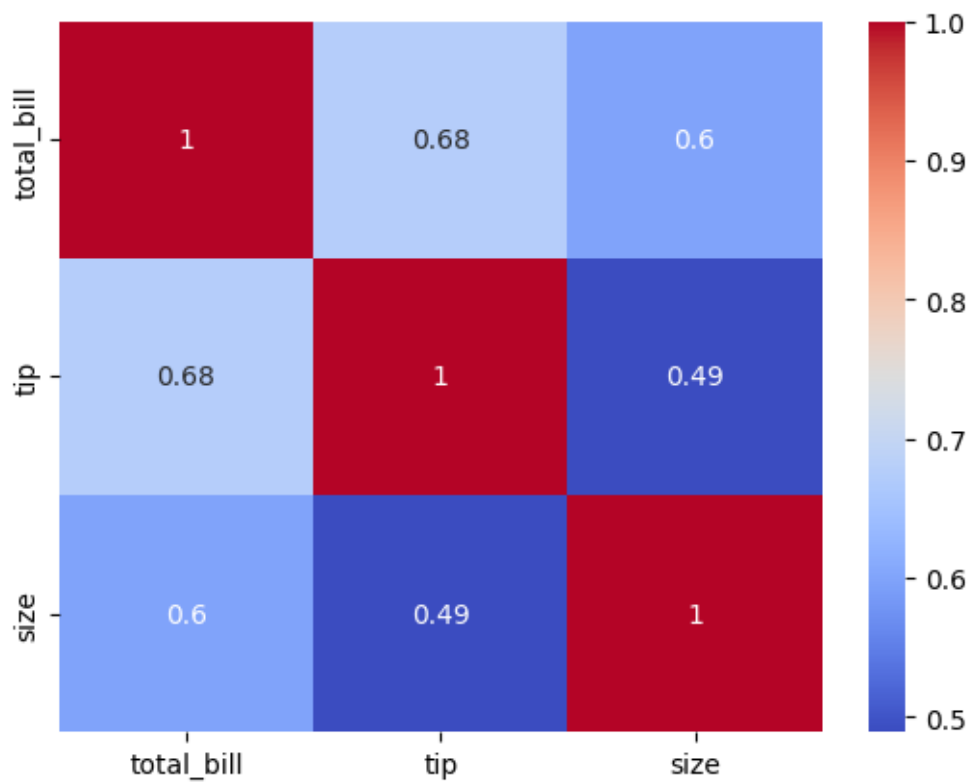
```
[38]: c
```

```
[38]:
```

	total_bill	tip	size
total_bill	1.000000	0.675734	0.598315
tip	0.675734	1.000000	0.489299
size	0.598315	0.489299	1.000000

```
[47]: sns.heatmap(c, annot=True, cmap="coolwarm")
```

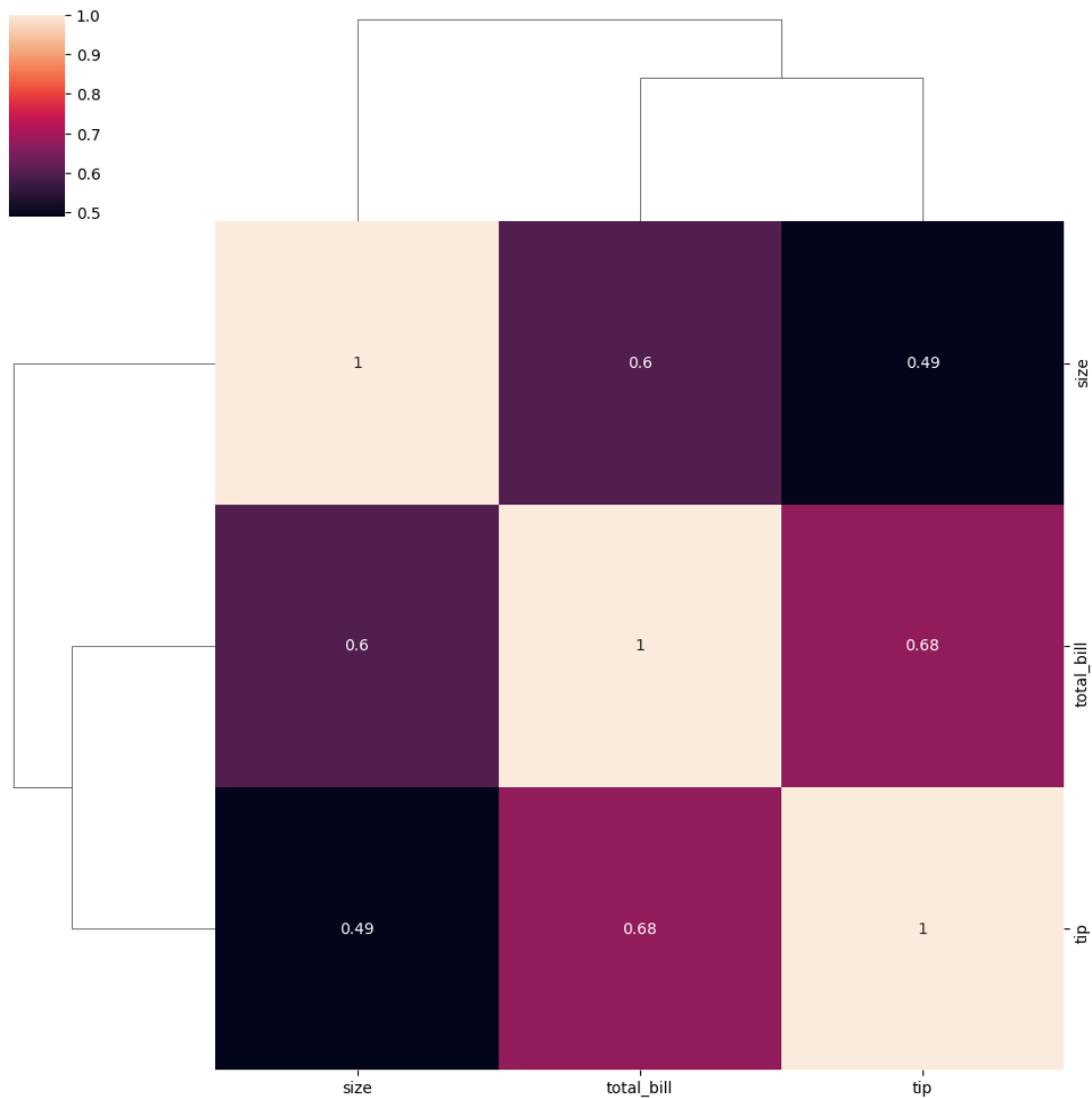
```
[47]: <Axes: >
```



```
[43]: # 0--1 ---->0
      # -3---+3----->0
```

```
[45]: sns.clustermap(c, annot=True)
```

```
[45]: <seaborn.matrix.ClusterGrid at 0x1ed0bc1ba10>
```



```
[46]: # sns.scatterplot()
```

```
[48]: sns.load_dataset("tips")
```

```
[48]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
..	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2

241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

[244 rows x 7 columns]

```
[49]: sns.load_dataset("flights")
```

```
[49]:
```

	year	month	passengers
0	1949	Jan	112
1	1949	Feb	118
2	1949	Mar	132
3	1949	Apr	129
4	1949	May	121
..	...	...	...
139	1960	Aug	606
140	1960	Sep	508
141	1960	Oct	461
142	1960	Nov	390
143	1960	Dec	432

[144 rows x 3 columns]

```
[51]: df2=sns.load_dataset("titanic")
```

```
[52]: df2
```

```
[52]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	
..	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	
887	1	1	female	19.0	0	0	30.0000	S	First	
888	0	3	female	NaN	1	2	23.4500	S	Third	
889	1	1	male	26.0	0	0	30.0000	C	First	
890	0	3	male	32.0	0	0	7.7500	Q	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True
..	...	...	...	...	...	...

886	man	True	NaN	Southampton	no	True
887	woman	False	B	Southampton	yes	True
888	woman	False	NaN	Southampton	no	False
889	man	True	C	Cherbourg	yes	True
890	man	True	NaN	Queenstown	no	True

[891 rows x 15 columns]

```
[53]: df2.isna().sum()
```

```
[53]: survived      0
      pclass        0
      sex           0
      age          177
      sibsp         0
      parch         0
      fare          0
      embarked      2
      class         0
      who           0
      adult_male     0
      deck          688
      embark_town    2
      alive          0
      alone          0
      dtype: int64
```

```
[54]: c= df2.corr(numeric_only=True)
```

```
[ ]: c
```

```
[ ]:
      survived      pclass      age      sibsp      parch      fare \
survived    1.000000 -0.338481 -0.077221 -0.035322  0.081629  0.257307
pclass      -0.338481  1.000000 -0.369226  0.083081  0.018443 -0.549500
age         -0.077221 -0.369226  1.000000 -0.308247 -0.189119  0.096067
sibsp       -0.035322  0.083081 -0.308247  1.000000  0.414838  0.159651
parch       0.081629  0.018443 -0.189119  0.414838  1.000000  0.216225
fare        0.257307 -0.549500  0.096067  0.159651  0.216225  1.000000
adult_male -0.557080  0.094035  0.280328 -0.253586 -0.349943 -0.182024
alone      -0.203367  0.135207  0.198270 -0.584471 -0.583398 -0.271832

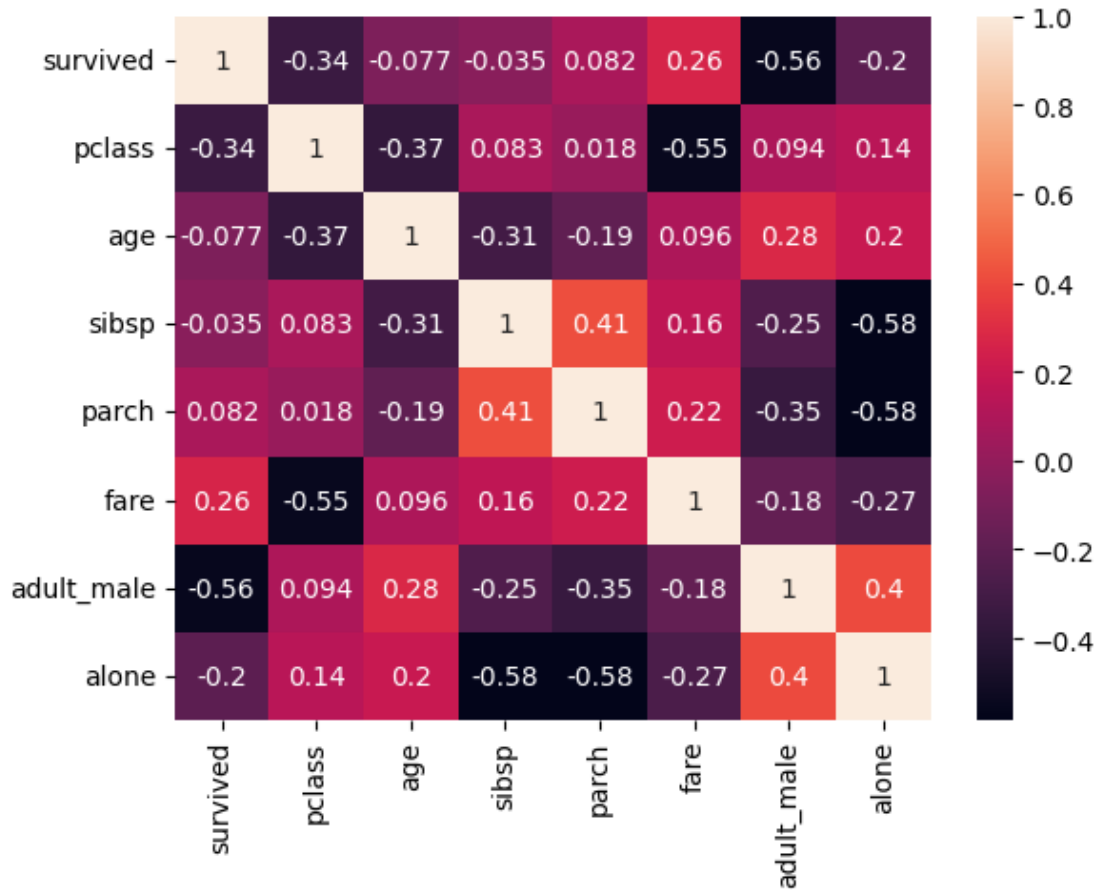
      adult_male      alone
survived    -0.557080 -0.203367
pclass       0.094035  0.135207
age          0.280328  0.198270
sibsp       -0.253586 -0.584471
parch       -0.349943 -0.583398
```



```
fare          -0.182024 -0.271832
adult_male    1.000000  0.404744
alone         0.404744  1.000000
```

```
[57]: sns.heatmap(c, annot= True)
```

```
[57]: <Axes: >
```



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[ ]:
```